clinical failure: random-effects meta-analysis (odds ratio)

Comparison: Intervention / Control

Methods

Odds ratios were pooled using a random-effects model with the DerSimonian–Laird estimator of between-study variance. Confidence intervals (95%) used the normal approximation. A continuity correction of 0.5 was applied to studies with any zero cell. 2 studies with no events in either arm were retained with this correction. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

19 studies were included (1097 participants in Intervention, 1080 in Control). The pooled odds ratio was 0.77 (95% CI [0.48, 1.22], z = -1.11, p = .267). The confidence interval includes 1, so these data do not show a clear difference between Intervention and Control. Heterogeneity: Q = 33.36, df = 18, p = .015. I^2 = 46.0%, which may represent moderate heterogeneity (Cochrane Handbook ranges). Between-study variance was τ^2 = 0.38. The 95% prediction interval was 0.21 to 2.80.

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Odds ratio	0.77	0.48 to 1.22	0.24	-1.11	.267

Table 2. Model fit

Statistic	Value
Studies (k)	19
Participants in Intervention	1097
Participants in Control	1080
Cochran’s Q	33.36
Heterogeneity degrees of freedom	18
Heterogeneity p	.015
I^2	46.0%
Between-study variance (τ^2)	0.38
95% prediction interval	0.21 to 2.80

Table 3. Study results

Study	Effect	95% CI	Weight
Gonzalez	0.57	0.17 to 1.92	7.3%
Prins	0.44	0.11 to 1.77	6.2%
Giamarellou	0.08	0.004 to 1.44	2.1%
Maller	1.40	0.67 to 2.94	10.7%
Sturm	0.19	0.0088 to 4.08	2.0%
Marik	0.39	0.23 to 0.68	12.2%
Muijsken	1.61	0.49 to 5.29	7.5%
Vigano	2.88	0.12 to 71.82	1.8%
Hansen	0.14	0.03 to 0.70	5.3%
De Vries	1.97	0.48 to 8.19	6.1%
Mauracher	0.07	0.004 to 1.31	2.2%
Nordstrom	0.66	0.13 to 3.28	5.3%
Rozdzinski	1.02	0.46 to 2.24	10.3%
Ter Braak	1.36	0.50 to 3.67	8.7%
Tulkens	1.00	0.02 to 52.98	1.2%
Van der Auwera	0.19	0.0086 to 4.06	2.0%
Klastersky	5.76	1.36 to 24.36	6.0%

Vanhaeverbeek	1.05	0.02 to 55.63	1.2%
Hollender	0.32	0.01 to 7.93	1.8%

clinical failure · Binary · Odds ratio · Comparison: Intervention / Control

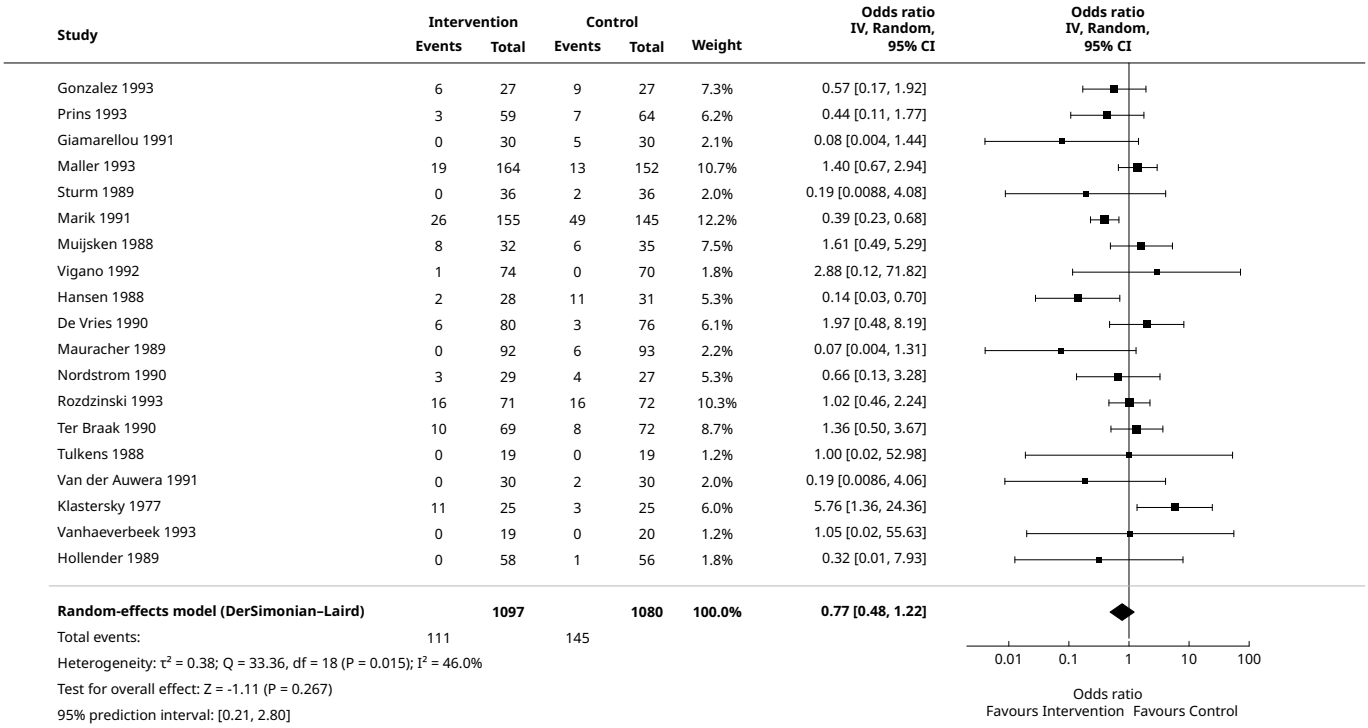

Figure 1. Forest plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: 0.77 Lower bound (95% CI): 0.48 Upper bound (95% CI): 1.22 p-value: 0.267
Summary — Heterogeneity	Q: 33.36 df: 18 p-value: 0.015 I^2: 46.0% τ^2: 0.38
Summary — Test for overall effect	Z: -1.11 p-value: 0.267
Summary — Prediction interval	Lower bound (95% PI): 0.21 Upper bound (95% PI): 2.80
Summary — Calculation scale (log)	Estimate: -0.26 Lower bound (95% CI): -0.72 Upper bound (95% CI): 0.20 SE (log scale): 0.236
Weights — Study weights	Study: Gonzalez Effect: 0.57 Lower bound: 0.17 Upper bound: 1.92 Weight (%): 7.3%
Weights — Study weights	Study: Prins Effect: 0.44 Lower bound: 0.11 Upper bound: 1.77 Weight (%): 6.2%
Weights — Study weights	Study: Giamarellou Effect: 0.08 Lower bound: 0.004 Upper bound: 1.44 Weight (%): 2.1%
Weights — Study weights	Study: Maller Effect: 1.40 Lower bound: 0.67 Upper bound: 2.94 Weight (%): 10.7%
Weights — Study weights	Study: Sturm Effect: 0.19 Lower bound: 0.0088 Upper bound: 4.08 Weight (%): 2.0%
Weights — Study weights	Study: Marik Effect: 0.39 Lower bound: 0.23 Upper bound: 0.68 Weight (%): 12.2%
Weights — Study weights	Study: Muijsken Effect: 1.61 Lower bound: 0.49 Upper bound: 5.29 Weight (%): 7.5%
Weights — Study weights	Study: Vigano Effect: 2.88 Lower bound: 0.12 Upper bound: 71.82 Weight (%): 1.8%
Weights — Study weights	Study: Hansen Effect: 0.14 Lower bound: 0.03 Upper bound: 0.70 Weight (%): 5.3%

Weights — Study weights	Study: De Vries Effect: 1.97 Lower bound: 0.48 Upper bound: 8.19 Weight (%): 6.1%
Weights — Study weights	Study: Mauracher Effect: 0.07 Lower bound: 0.004 Upper bound: 1.31 Weight (%): 2.2%
Weights — Study weights	Study: Nordstrom Effect: 0.66 Lower bound: 0.13 Upper bound: 3.28 Weight (%): 5.3%
Weights — Study weights	Study: Rozdzinski Effect: 1.02 Lower bound: 0.46 Upper bound: 2.24 Weight (%): 10.3%
Weights — Study weights	Study: Ter Braak Effect: 1.36 Lower bound: 0.50 Upper bound: 3.67 Weight (%): 8.7%
Weights — Study weights	Study: Tulkens Effect: 1.00 Lower bound: 0.02 Upper bound: 52.98 Weight (%): 1.2%
Weights — Study weights	Study: Van der Auwera Effect: 0.19 Lower bound: 0.0086 Upper bound: 4.06 Weight (%): 2.0%
Weights — Study weights	Study: Klastersky Effect: 5.76 Lower bound: 1.36 Upper bound: 24.36 Weight (%): 6.0%
Weights — Study weights	Study: Vanhaeverbeek Effect: 1.05 Lower bound: 0.02 Upper bound: 55.63 Weight (%): 1.2%
Weights — Study weights	Study: Hollender Effect: 0.32 Lower bound: 0.01 Upper bound: 7.93 Weight (%): 1.8%
Inference method	Normal approximation

Pooled Odds ratio: 0.77 (95% CI 0.48 to 1.22).

95% prediction interval: the underlying effect in a new study; this is not the confidence interval of the pooled estimate.

References

1. Random-effects meta-analysis: DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. Controlled Clinical Trials, 7(3), 177-188. doi:10.1016/0197-2456(86)90046-2.

2. Random-effects meta-analysis: Hedges, L. V. (1983). A random effects model for effect sizes. Psychological Bulletin, 93(2), 388-395. doi:10.1037/0033-2909.93.2.388.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.